Formal Modeling of the Social Impact theory using the example of effort matching

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Author Note

Author roles were classified using the Contributor Role Taxonomy (CRediT; https://credit.niso.org/) as follows: Samuel Riess: writing, conceptualization

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Abstract

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Objective

In this report, we formalize an existing psychological theory to test the relation between the emerging phenomena and the corresponding data by means of a computational model, based on the underlying theory. Following the methodological framework outlined by Borsboom (Borsboom et al., 2021), we generate simulated data for a fictional experiment inspired by the original study in which the phenomenon was first described. We are mainly interested in replicating key empirical findings associated with the phenomenon and to demonstrate that the model is capable of reproducing the phenomenon across different hypothetical scenarios. The need for such work is grounded in broader concerns within psychological science, known under terms like "reproducibility crisis" (citation Borsboom). Hence, our aim is to take a closer look at the relationship between the underlying theory and the emerging phenomenon, evaluating how well the theoretical assumptions can account for the observed effects.

Introduction of the theory

We selected our social psychology theory because it provides a framework of clearly operationalized variables, allowing a precise measurement for input factors and their resulting phenomenon. The theory, named "social impact theory", was first mentioned in 1973, but the large framework with all assumed mechanisms and effects was published 8 years later (Latané, 1981). Between these publications, and chosen for our central phenomenon emerging from social impact, Latané published "Many Hands Make Light The Work: The Causes and Consequences of Social Loafing" in which the term "social loafing" was first defined. There, they tried to explain it using the first approaches of this theory. For adding more depth to the phenomenon we are referring to an extending mechanism, named "Effort Matching" (Jackson & Harkins, 1985).

Core constructs

Our formalization is grounded on some core constructs. These are orientated at quotes we withdrawed directly from the existing research. Some constructs were specified differently over the whole report, so we aimed to label them in a way that serves a greater interpretation.

Of course, these labels are our own understanding of the constructs, further researchers may focus on other elements. Furthermore, the scope is strongly limited to the main social loafing research, we only differ if a special need or cause occurred. The central social impact theory contains more than our chosen phenomenon so we were able to further limit the scope.

All constructs and their sources can be seen in (Table 1), our presented VAST model in Figure 2 contains them in a connected and graphic way to understand the underlying mechanism of the phenomenon. As one can see, the initial constructs that cause the effect are the Co-Targets (CT) and Sources (S) of social impact, which represent group members and e.g. experimenters who supervise the whole group in an experiment. Of course, the sources can be every kind of person that performs pressure or impact on a certain group or individual. On this note, it is worth mentioning that pressure and impact are not further separated so we're sticking with the definition "pressure" in this report, resulting in our construct P. The contingency between input and outcome (CON) marks a prior expectation, not an objective fact. That's why its presented as part of the "Psy"-Box. It leads later on to the "incentive to work equally hard" (INC), linked with a proportional relationship (Latané et al., 1979). Their values are the same, whilst INC contains a more motivational nature. The final psychological output of the mechanism is the individual effort (IE) which is the cause for the manifest output later on. The constructs "effort matching condition" (EMC) and "loafing expectation" consist both of three conditions, marking the 3 conditions of Effort matching we I implemented in the research from another source(Jackson & Harkins, 1985). We didn't include constructs like "identifiability"" due to our strict replication and setting identifiability to "not given", based on the original experiments (Latané et al., 1979). The same reason accounts for disregarding the constructs immediacy and strength of Source or Co-Targets.

Identification of the relevant Phenomena

In (Latané et al., 1979), the resulting social loafing is described as followed: "People exhibit a sizeable decrease in individual effort when performing in groups as compared to when they perform alone". The Effort Matching mechanism which underlies this phenomenon is defined in another research, stating: "they will maintain equity by matching the effort levels they expect from their partners" (Jackson & Harkins, 1985). This was already discussed

before (Latané et al., 1979) but further investigated in the newer report. The idea is that social loafing can be eliminated if the expectation to loaf, drawn to the effort of other group members, is capable of eliminating the central social loafing mechanism.

To create a link between this phenomenon and our formalization of the social impact theory, we used our visual tool once more. The resulting phenomenon itself can be seen in Figure 1.

To address the robustness of social loafing and therefore the very reason for investigating it further, we part it into two sections: Strenth of evidenve and generalizability, using the UTOS framework. We get our overview for both by analyzing a Meta-Analysis concerning social loafing (Karau & Williams, 1993) which conducts about 78 studies of the topic. We found quotes like “using a range of subject populations varying in age, gender, and culture. This leads to a variation of Units (u). The variation of treatment (T) is supported by using different attributes of the task itself, e.g. complexity, type of effort, quantity or quality. Within- and between-subjects comparisons across coactive and collective conditions took place in investigating social loafing. the variation of Outcomes (O) is supported by operationalizing the dependent variable through cognitive of physical effort. sometimes, it was based on an individual or group level. Finally, the (S) of the experiments took place in different coutries, what we interpreted as a generalization over different cultures and a confirmation of the last UTOS-dimension. The strength of evidence can be acknowledged with mean effect sizes that”differed significantly from the 0.00 value that indicates exactly no difference”. The most pithy evidence can be found in the result section. Directly, the mean effect size was found to be $d = .44$ which was statistically significant ($p < .001$), presenting a moderate effect according to Cohen’s general conventions. Taken together, the phenomenon can be considered robust because the whole UTOS framework and the strength of evidence are supported with empirical data.

Method

All documents used in the process of writing this report can be found and downloaded from the GitHub repository: https://github.com/SamRiess/SocialLoafing_FINAL.git This report refers to version number v1.0.0. See the .cff file for further information.

used software

Throughout our formalization, we use the Visual Argument Structure Tool (VAST, (Leising et al., 2023)) to build a graphic visualization of the theory and resulting mechanisms. With this methodology, we are able to define relationships, our understanding of core constructs and the mechanisms that take place in the process of social loafing and effort matching. For the computation of our simulated Data, we used R(hier Versionsnummer einfügen). Our graphic representation of the formalized theory was enabled throughout the application draw.io in which we created our VAST model.

Formalization of the (Proto-)Theory

We used the VAST (Visual Argument Structure Tool) as a graphic form of formalization for our theory. Further information about the VAST and its application can be found in Leising et al. (2023). In the process, it was crucial to limit our scope, referring to both theory and phenomenon. The main goal should be understandability, not the whole integrity of the full theory. We were limiting our scope to the following aspects: we set the identifiability of the individual effort to non-existent. This can be justified for that the identifiability wasn't part of the experiments for social loafing (Latané et al., 1979), neither was it for effort matching (Jackson & Harkins, 1985). Further, we ignore the mentioned multiplicative connection between immediacy, strength and number of Co-Targets or Sources (Latané, 1981). These weren't, again, part of previous research so we didn't focus on that specific part of social impact theory. A few more additional assumptions were crucial for our finalized theory and model to work. Other mentioned constructs like the default expectation for other peoples loafing wasn't specifically evaluated and included. We ought to deal with this in using the mechanism in the Psy box to generate this default expectation, tho it wasn't defined directly. With this approach, the underlying process gains more understandability while the high and low Effort condition could be included into the Superfunction for our psyche process without complicating the model itself. The extension of the maximum capacity was necessary for simulation purposes, it isn't mentioned in the theory itself either. Without implementing this construct, we weren't able to compute a manifest Variable cause the main output of the psychological process is just a percentage number which cannot be measured

without a baseline which presents 100% of the possible effort. See Table 2 for further values. As we're introducing later in this report, these restrictions or extensions weren't loss-making for our Model and replication of the loafing effect and therefore providing a post-hoc evaluation. Our resulting prototype theory which ought to explain the phenomenon is presented in (Figure 2). The "relationships" section shows each relationship and its justification, based on quotes from previous research (labeled with a "q" in Table 3) or our own assumptions or restrictions. In keeping this section short, I am mixing up the explanation of the model with its justification of the described relationships.

Variables

All our used variables for each of the constructs, if shown in our VAST, can be seen in (Tabelle einfügen).

Relationships

All relationships are mentioned in Table 3 which the relevant quotes, labeled with a "q" in the following section. The relationship between CT, S and P is presented in Figure 3. We aimed to write a function that is consistent with exact citations from the sources. Originally, the pressure can be lowered throughout the number of CT but increased with more S. As we can see in the graphs, the higher the Number of S is, the higher gets the felt impact on one subject. This represents the classic increase which is mentioned in q4. But with an increase of the number of Co-Targets, the impact lowered rapidly. Therefore, consistent with the general interpretation of social impact theory, felt impact is divided up among the group members (q5-q8). No specific expectation about the weight of each process is given, so we filled it up with our own values.

The left path of our VAST (Figure 2) presents the main mechanism which produces the phenomenon. It starts with the relationship between CT and CON which was rather intuitive due to the specific function derived from the core report, verified with (Latané et al., 1979) and specifically in q1-q3. It is presented in Figure 4.

After that, the transformation into INC can be represented by a proportional increase: INC gets the same values as the previously computed CON (Figure 5). This was not fully based on a focal theory, but we infered that an incentive to give one's own full effort directly

results from the depicted CON, also supported with q14.

Things were getting more interesting with the function that computes the resulting IE, using all previously computed and simulated input variables. Both paths of our own VAST meet together in this relationship. The plot can be viewed in Figure 6, showing the first influence of our EMC as well. I wrote a function which transforms the manifest EMC into the latent construct LE, named t3 and supported by q12/q13. If the subject takes action in the LE (low effort) or HE (high effort) condition, effort matching takes place and fixes individual effort to a previously fixed value for each condition. These two values for effort matching conditions can be chosen individually, but I made sure that its higher in the HE condition compared to the LE one. No further information about these values was found in the research, so this addition was necessary (Jackson & Harkins, 1985). The blue graph shows the natural social loafing mechanism for the individual effort, based on no information whatsoever about the others effort, used for the replication condition SLR. If the pressure is high, the function starts higher cause the subject feels more bound to the outcome. Secondly, with more INC the subject is getting higher values in the effort. We argue that, besides great pressure, the effort won't reach its maximum if the INC isn't maximized. The basic idea of the relationship is supported by q9-q11. We had to adjust the function to fit in our theory and model so P and IE had a range from 0-1.

The simulation-relevant relationship we have to show is the transformation of IE and MC into the IO. As can be seen in Figure 7, the more MC each subject has, the higher the value for IO is. That occurs because we defined IE as a percentage of 1 (e.g. 0.6 = 60%) which is offset against the MC. Hence, with more IE, IO is increasing simultaneously. We concluded a interaction effect between MC and IE that accounts for the final value of one's own output. The transformation function was needed to compute the manifest outcome for each person (e.g. in q15) and the group means for our statistical analysis later on.

The behavior of the combined system is shown in (). It is pretty similar to the IE function but deviates in its own input variables. In this case, we used all manifest ones and threw them into the whole psychological process that it labeled "Psyche" (see Figure 1 for clarification). The resulting "superfunction" can be described as followed:

Evaluation of the formal model

Finally, we got to test our model with our simulation. We simulated the original experiment for social loafing, using the sound pressure as the individual outcome (**?**) but added the EMC predictor, known in our theory by "Effort matching condition". With a sample size of n=36 (view (**?**), we were able to test for the original effect and the effort matching mechanism in one sample). The descriptives for both analyses are showcased in Figure 8 and Figure 9. These are highly influenced by the noise term of p3 and can be adjusted freely. We chose the parameters mean = 0 and sdd = 0.3 for our "noise term" of that function, cause we assume that variability comes into the model throughout many sources: e.g. fatigue, personality, or the current mental state.

Due to our 2 computed ANOVAS which rely on the same sample, we needed to adjust our p-value with the Bonferroni method. Thus, the new α-niveau is divided by 2 and comes down to .025. It turns out that both effects, on one side the social loafing but on the other the elimination of this phenomenon throughout implementing a low and high Effort condition were supported by our present simulation and therefore our model. In proving the results, we fall back on the original analysis (Latané et al., 1979) for which all resulting graphs can be viewed in Figure 10. The main replication was significant for the number of Co-Targets had a significant effect on the individual sound output $F(2, 105) = 18.56$, $MSE = 11.78$, $p < .001$, $\hat{\eta}^2_G = .261$. This is consistent with the findings of (Latané et al., 1979) and proves that our model is capable of producing the main phenomenon. Whilst replicating the findings of effort matching (Jackson & Harkins, 1985), we compared alle three Pairs of "Alone vs. Group", each for one condition of EMC. Only in the SLR-condition, we detected a significant decrease in the outcome between the group and alone situation (p < .001). For the low effort condition the difference was no longer significant with p=.372, equal results appeared in the the high effort condition with p=.175. Therefore, we've shown that Effort Matching emerged via our model and is consistent with the literature in eliminating the social loafing effect thoroughly holding the loafing expectation constant.

Meta-reflection

Some issues need to be discussed at the end. First, they original paper contained many jangle-fallacies, so it was quite difficult to form clear definitions for our main constructs. An alternative VAST image (Figure 11) illustrates this conceptual separation, highlighting that each construct should be clearly defined without redundant or overlapping terminology. Second, it was a challenge to implement the sources (S) into our model, as these factors were conceptualized in the underlying theory but not explicitly manipulated within the experimental design (Latané et al., 1979). A good limitation of the scope was downright necessary in formalizing this theory while using VAST. Hence, the mentioned inconsistencies in the original experiment and its explanation had to be managed. In regard of the simulation, the decision if the VAST itself was to contain a group level was partially difficult - in the end we decided to stick with the individual output as the final construct. For using a group level directly in the VAST we had to adjust the simulation, the Co-Target construct to "group members" and therefore the whole psychological mechanism of social loafing.

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Appendix

Concept Table - Social Loafing
Editor: Samuel Riess

This document contains the construct, relationship and variable table for the Report "insert Title here" for the 2026 FOMO Course, specifically for the report of Samuel Riess.

The repository for the report is linked here:
https://github.com/SamRiess/SocialLoafing_FINAL.git

Table 1

Table of core constructs

ID	Type - Phenomenon - Concept - HOC	Short name	Quote	Reference	Rel. type (n, p, i, r, ...)	Comment (e.g., identified gaps & inconsistencies)	Incl. (Y/N)?
CT	C	Co-Targets	-"The more people who are the target of this pressure, [...]" (S.823) - "According to Latane's (1981) social impact theory, people can be viewed as […] targets of social impact." (S. 682) AND - "and number of group members" (Latané et al., 1979, S. 823) - "impact is divided among the group members" (Latané et al., 1979, S. 830) - "However, if a person is a member of a group that is the target of social forces from outside the group" (Latané et al., 1979, S. 823)	Latané et al., S.823 (1979) AND Karau & Williams, S. 682 (1993)	n1, p1,p2, n12		Y

S	C	Source of pressure	- “The impact that the experimenters have on an individual [...]” (S.825) - “According to Latane's (1981) social impact theory, people can be viewed as either sources […] of social impact.” (S. 682)	Latané et al., S.825 (1979) AND Karau & Williams, S. 682 (1993)	n2, p2	these persons are not described as a pressure precisely, but we can assume that they are the reason why the pressure occurs we set: pressure = impact	Y
CON	C	Contingency between input and outcome	“[...] the contingency between their input and the outcome [...]” (S. 830)	Latané et al., S. 830 (1979)	p1, n3, t1	here, contingency is simply explained through the word itself, no further definition	Y
INC	C	Incentive	“[...] incentive to work equally hard in groups of different sizes.” (S.830)	Latané et al., S. 830 (1979)	t1, n4, p3		Y
P	C	impact/ pressure	“the amount of social pressure on the target person” (S. 823) “pressures on each individual” (S. 830) / “In a group performance situation in which pressures to work come from outside the group” (S. 830)	Latané et al., S.823 (1979)	p2, n5, n6, p3	impact and pressure are no really distinctive concepts, cause impact can officiate as a social	Y

			AND “[...]the impact of these forces on any given member [...]” (S.823)			facilitation. here, we assume its the same cause we limited the scope to the negative form of social impact, no facilitation at all	
IE	C	individual effort	“individual effort” (S. 822 ff.)	Latané et al., S.822 ff. (1979)	p3, n7, t2	In this paper, there is no real distinction between “effort” and “performance”. These concepts were used partially synonymously. The problem is, they were not defined.	Y
Psy	HOC	Psyche	no citation	/	n11	We defined this HOC as the space in which the mental process is defined	Y
IO	C	individual output	“in this case n times the average individual output” (S. 825)	Latané et al., S.825/S.	n13, n14, n15, t2		

			AND “Since individual scores” (S. 830) AND - “leading to an apparent drop in individual performance, a phenomenon we call social loafing” (S. 825) - “sum of their individual performances” (S. 829)	829/S.830 (1979)			
MC	C	maximal capacity	“dyads pulled at 91% of the sum of their individual **capacities**, trios at 82%, and groups of six at only 78%” (S. 825)	Latané et al., S.825 (1979)	n16, t2	maximal capacity of individual, when 100 sources and 0 cotargets are present (no citations are available, this is our own interpretation to simulate the phenomenon)	Y
ME	HOC	Matching of effort	“this effort matching interpretation” (S.1204) “they will maintain equity by matching the effort levels they expect from their partners” (S.1200) “social loafing occurs because individ uals expect others to slack off in groups and, therefore, reduce their own efforts to	Harkins & Jackson, S.1200/1204 (1985) AND Karau & Williams, S. 683 (1993)	/	this effect occurs only when the scope is limited to holding identifiability constant throughout the experiment	N

			maintain equity” (S.683)				
EMC	C	Effort condition	“In the HE condition, she said she had tried very hard on the practice trial and, because she thought the research was interesting, she was going to try hard throughout the experiment; in the LE condition, she said she had not tried on the practice shout and, because she thought the research was boring, she wasn't going to try hard during the rest of the experiment; and in the SLR condition, she simply made small talk until the experimenter returned” (S.1202)	Harkins & Jackson, S.1202 (1985)	n17, t3	this is purely a condition construct, no explicit values were give in Harkins & Jackson (1985), manifest variable outside the Psy-Box	Y
LE	C	loafing expectation	“expectations of the amount of partner effort” (S.1204) AND “expectations concerning the partner’s intended levels of effort”	Harkings & Jackson, S.1204 (1985)	n18, t3, p3	no values were given here either, but this is strictly a latent variable - we transform it from the effort matching condition to have a construct that matches the condition in the Psy-Box	Y
SL	P	Social loafing	“people exhibit a sizeable decrease in individual effort when performing in groups as compared to when they perform alone”	Latané et al., S.822 (1979)	i1	this is the core phenomenon for the report	

Table 2

Table of variables.

All concepts that are not higher-order-concepts are listed here. Higher-order concepts typically aren't reducible to a single metric and are therefore not compromised into one single variable.

Construct	Source (if given)	Scale level	Range/Values	Anchors
CT	"and *number* of group members" (Latané et al., 1979, S. 823)	discrete	$\in \mathbb{N}_0$	0 = "zero cotargets are present"; n = "number of cotargets (other persons in the evaluated group, number of group members minus 1)"
S	"*number* of these other persons." (Latané et al., 1979, S. 823) AND "Thus, as the *number of performers decreased, the size of the audience increased*. According to Cottrell's evaluation apprehension hypothesis (1972), the presence of an evaluative audience should enhance performance for a simple, well learned task such as rope pulling,	discrete	$\in \mathbb{N}_0$	"0 = "zero persons that cause the social pressure"; n = "number of persons that cause the pressure"

	and, although there is little supportive evidence, it seems reasonable that the *larger the audience*, the greater the enhancement“ (Latané et al., 1979, S. 826)			
P		continuous	∈ [0; 1[	0 = “zero persons are source of social pressure”, “no social pressure” 1 = “maximal social pressure (a theoretical amount on infinite spectators are present)”
CON	“Individuals could "hide in the crowd" (Davis, 1969) and avoid the negative consequences of slacking off, or they may have felt "lost in the crowd" and unable to obtain their fair share of the positive consequences for working hard.” (Latané et. al, 1979, S. 830) “reward” (Latané et. al, 1979, S. 830)	continuous	∈ [0; 1]	0 = “expectation that the share (reward/punishment) of each member is completely independent of what they’ve given” 1 = “expectation that their share (reward/punishment) is equal to what they’ve given”
INC		continuous	∈ [0; 1]	0 = “no incentive to give any individual effort” ;

				1 = “incentive to give full effort”
IE	e. g. “dyads pulled at 91% of the sum of their individual capacities, trios at 82%, and groups of six at only 78%” (Latané, 1979, S. 825)	continuous	$\in [0; 1]$	0 = “the person shows no effort at all to bring in some contribution to the group task”, 1 = “the person shows their maximal effort in contributing to the group task”
O	“sum of their individual capacity” (Latané, 1979, S. 824); e. g. “dyads pulled at 91% of the sum of their individual capacities, trios at 82%, and groups of six at only 78%” (Latané, 1979, S. 825)	continuous	$\in [0; 1]$	0 = “maximal social loafing effect, nobody does anything”; 1 = “sum of the maximal individual performances” “no social loafing effect” “everybody gives their best” o = “percentage of the theoretical maximal outcome (sum of each individual capacity) of the whole group”
IO	“dyads pulled at 91% of the sum of their individual capacities, trios at 82%, and groups of six at only 78%” (Latané, 1979, S. 825)	continuous	$\in [0; \inf[$	0 = dynes/cm^2 n = dynes/cm^2
MC		continuous	$\in \mathbb{R}_0$	depends on task (clapping with hands, rope pulling)

EMC	“she said she had tried very hard on the practice trial and, because she thought the research was interesting, she was going to try hard throughout the experiment” (Jackson & Harkins, 1985, S.1202) AND “she said she had not tried on the practice shout and, because she thought the research was boring, she wasn’t going to try hard during the rest of the experiment” (Jackson & Harkins, 1985, S.1202) AND “she simply made small talk until the experimenter returned” (Jackson & Harkins, 1985, S.1202)	ordinal	1, 2 or 3 is possible	1 = low effort condition, 2 = Social loafing replication condition, 3 = high effort condition
LE	view “EMC” for citations, they’re identical	ordinal	“match_low”, “no_info”, “match_high”	“match_low” = the subject thinks the Co-Targets were giving their full effort in the practice trial, “no_info” = the subjects has no information whatsoever about

			the others goal in effort, “match_high” = the subject thinks the Co-Targets had not tried hard in the practice trial	

Table 3

Table of relationships.

Each variable in the VAST to which a one-sided arrow is pointing is a dependent (endogenous) variable. For each endogenous variable, there is provided a mathematical function. It takes all incoming arrows as input variables.

Note: The relationship p3 and t2 contain a R Code and not simply a mathematical relation. This is based on the nature of the VAST and the extension with the Variable LE and EMC.

Output variable	**Function to compute the output variable**	**Comment**	**To what extent derived from theory?[a]**	**To what extent empirically backed?**
relationship P1 => CON	CON = 1/(CT+1) => CON = 1/n	quote q3 mentions the “reward” but if each group member performed up to its own capacity would feed an inconsistency between the effort (100%) and the reward/outcome (1/n)	CT → CON: q1: “We believe that social loafing occurs because there is usually a stronger perceived contingency between individual effort and valued outcomes when working individually. When working collectively, factors other than the individual's effort frequently determine performance, and valued outcomes are often **divided among all of the group members**.“ Karau & Williams, S. 684 (1993) q2: “which both suggest that individuals' outcomes are more contingent on their efforts when working with others who are expected to perform poorly.“ Karau & Williams, S. 696 (1993)	

		q3: “the reward is psychologically divided equally among the participants, each getting 1/n units of reward” (Latané, 1979, S.830)
relationship P2 => P	P = S/(S+CT+1)	q4:“[...]the amount of social pressure on the target person should increase as a multiplicative function of the strength, immediacy, and number of these other persons.” (Latané, 1979, S.823) => derived from theory q5: “Impact is divided up among the group members, in much the same way that responsibility for helping seems to be divided among witnesses to an emergency” (Latané, 1979, S.823) => derived from theory q6: “If a person is the target of social forces, increasing the number of other persons also in the target group should diminish the pressures on each individual because the impact is divided among the group members.” (Latané, 1979, S. 830) CT → P: q7: “The more people who are the target of this pressure, the less pressure should be felt by any one person.” (Latané, 1979, S. 823) => derived from theory

			q8: “increased group size to result in reduced effort” (Latané, 1979, S. 823)

relationship P3 => IE	get_IE <- function(INC, P, LE) {
	  # the Social Loafing baseline (Latané)
	  # whilst CT increases, the output decreases
	  loafing_effort <- 0.5 * INC + 0.5 * P
	
	  # the effort matching logic
	  # we're definign fix values which represent the expectation for the "alone"-
	  # condition of other subjects (independent of the current group size)
	  final_IE <- case_when(
	  # condition: co-Target is not loafing in its "alone"-condition
	  LE == "match_high" ~ 0.9,
	
	  # condition: Co-Target is loafing in its "alone"-condition
	  LE == "match_low" ~ 0.4,
	
	  # In der no_info Bedingung greift das klassische Modell (Loafing)
	  LE == "no_info" ~ loafing_effort,
	
	  TRUE ~ loafing_effort
	  )
	
	  return(final_IE)
	}
		we added the effort matching condition/variable later on, no real mathematical relation could be found in the underlying paper
		
		this function includes one of two sources of variability between the subjects cause we added the noise term on top of it (but only in the superfunction version of it)
		
		For this function, it was reasonable to present the full R code.
			q9: “Latane further suggests that just as psychophysical reactions to external stimuli can be described in terms of a power law (Stevens, 1957), so also should reactions to social stimuli, but with an exponent having an absolute value less than 1, so that the nth person should have less effect than the (n - l)th.” (Latané, 1979, S. 823)
			=> derived from theory
			
			P → IE:
			q10: “The more people who are the target of this pressure, the less pressure should be felt by any one person. Since people are likely to work hard in proportion to the pressure they feel to do so, we should expect increased group size to result in reduced efforts on the part of individual group members.” (Latané, 1979, S. 823)
			
			q11: “If a person is the target of social forces, increasing the number of other persons also in the target group should diminish the pressures on each individual because the impact is divided among the group members. In a group performance situation in which pressures to work come from outside the group and individual outputs are not identifiable,

			this division of impact should lead each individual to work less hard. Thus, whether the subject is dividing up the amount of work he thinks should be performed or whether he is dividing up the amount of reward he expects to earn with his work, he should work less hard in groups." (Latané, 1979, S. 830) q12: "in fact, participants matched their own efforts to these new expectations" (Jackson & Harkins, 1985, S.1199) q13: "by matching the effort levels they expect from their partners" (Jackson & Harkins, 1985, S.1200) => derived from theory
relationship t1 => INC	INC = CON		q14: "Thus, under these assumptions, if his own performance cannot be individually monitored, an individual's incentive to perform should be proportional to l/n." (Latané, 1979, S. 830) => derived from theory
relationship t2 => IO	IO = MC * IE	since we need so compute a manifest outcome for each person and therefore for the experimental group means we had to multiply the percentage of individual effort with the maximum capacity	q15: "as with rope pulling, output can be measured in simple physical units that make up a ration scale" => derived from theory

relationship t3 => LE	get_LE <- function(EMC) {
	LE <- case_match(EMC,
	1 ~ "match_low",
	2 ~ "no_info",
	3 ~ "match_high"
	)
	return(LE)
	}
		we transformed the manifest construct/variable EMC into the latent construct/variable LE what created the need for such transformation	=> Not based on focal theory, but rather on common sense or other theories

---

Each resulting function in R was tested for functioning with vectors. This was crucial for guaranteeing a smooth simulation later on.

[a] Response scale:

- Dictated by theory (add IDs from the Construct Source Table as reference)
- Derived from theory
- Loosely inspired by theory
- Not based on focal theory, but rather on common sense or other theories
- Not at all

Figure 1

Structure of the VAST phenomenon

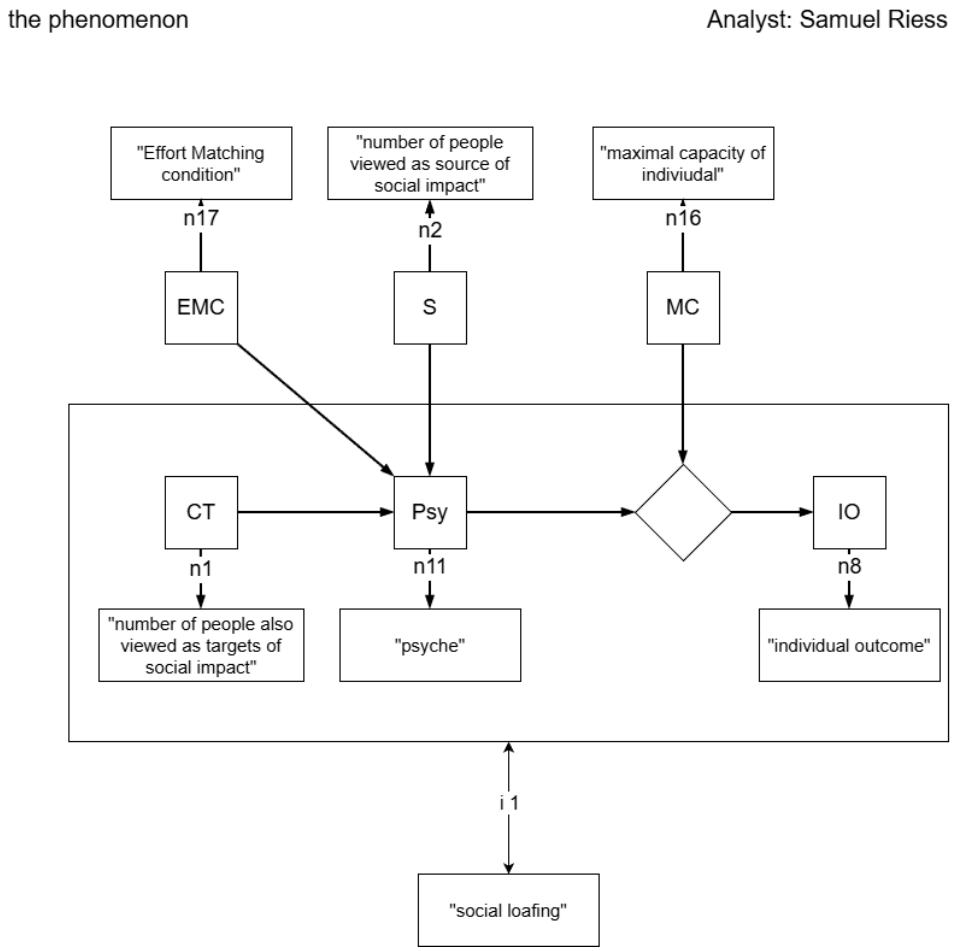


Figure 2

Structure of the VAST theory and mechanism

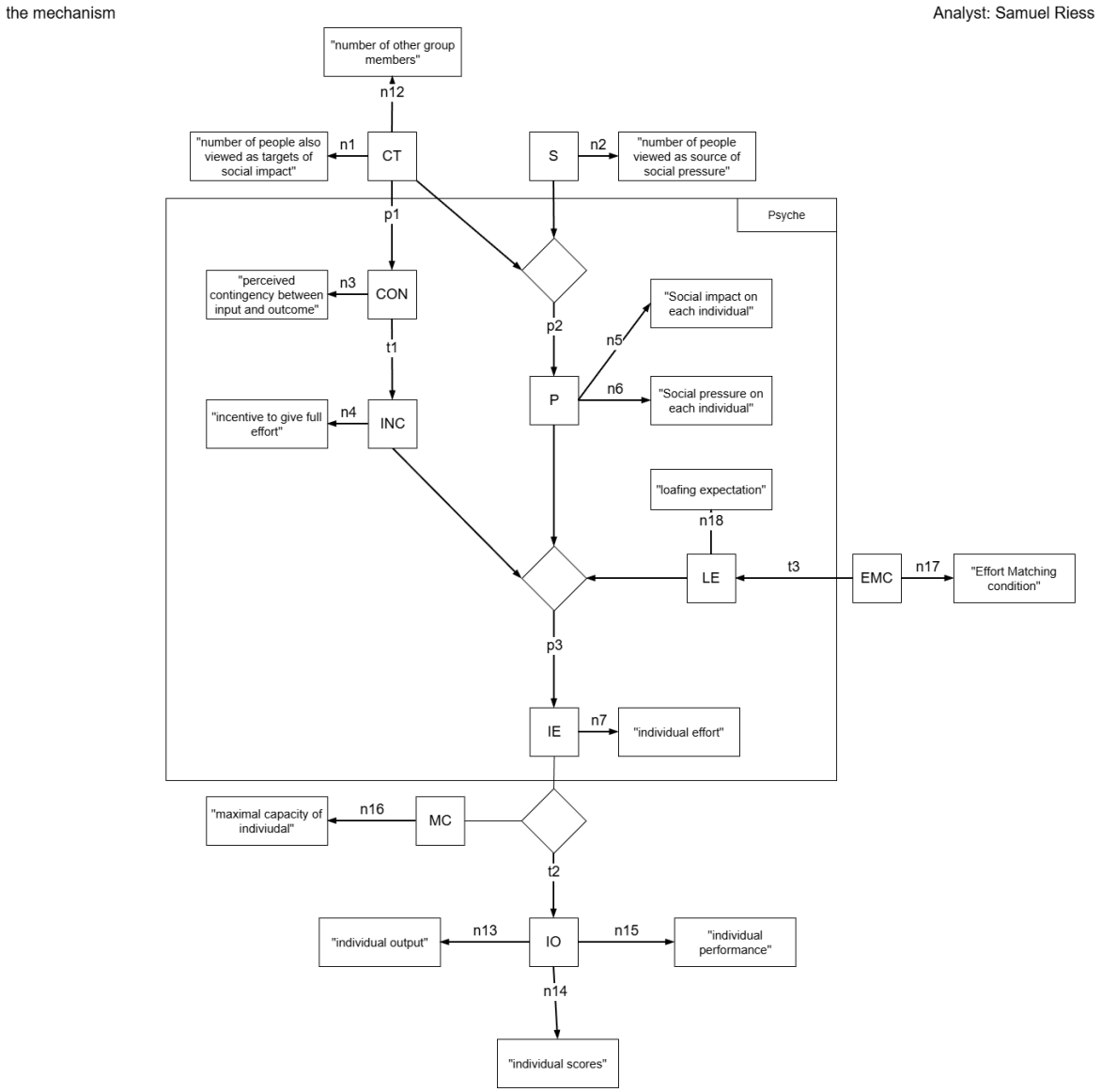


Figure 3

Plot for the relationsship p2

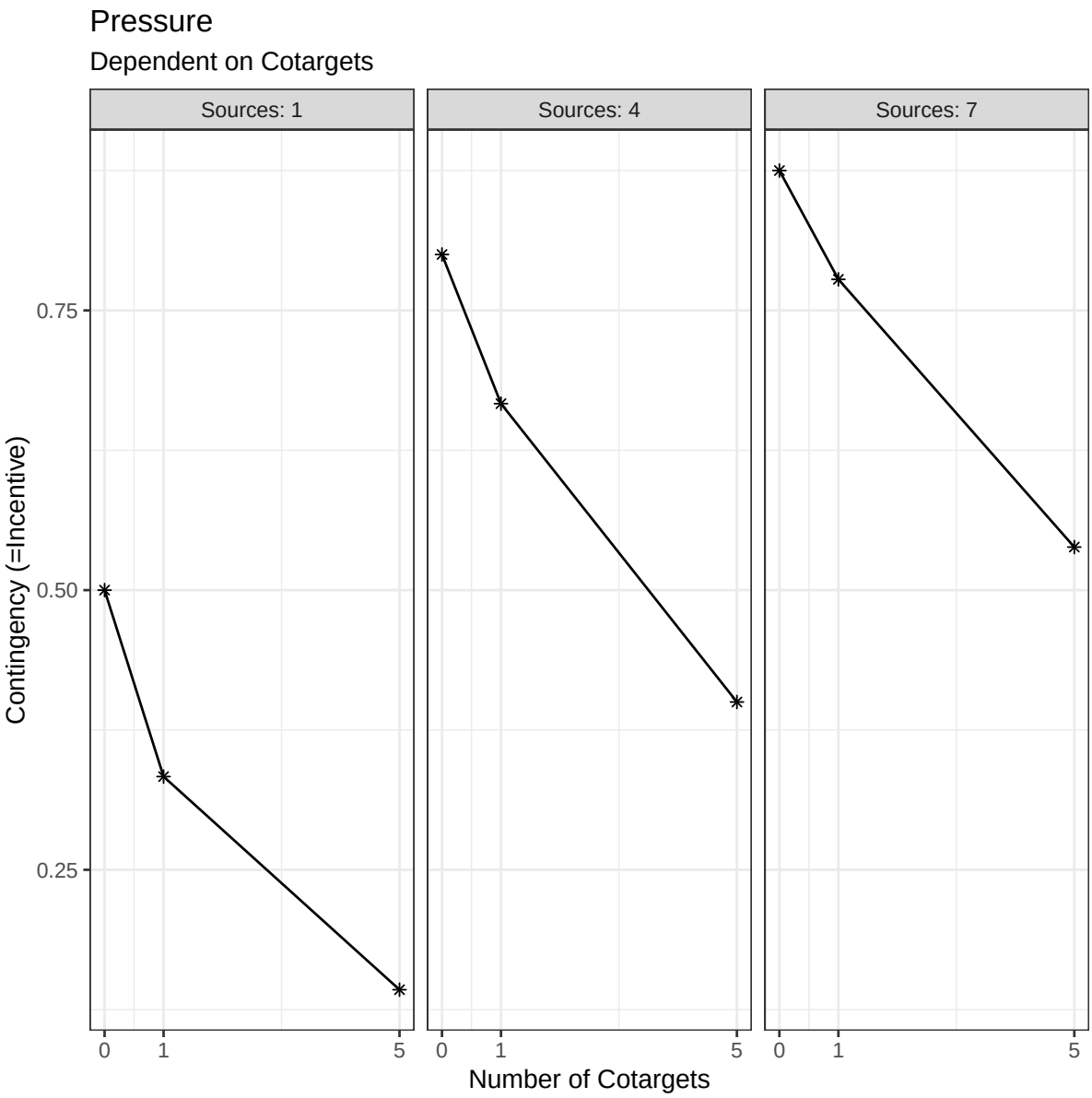


Figure 4

Plot for the relationsship p1

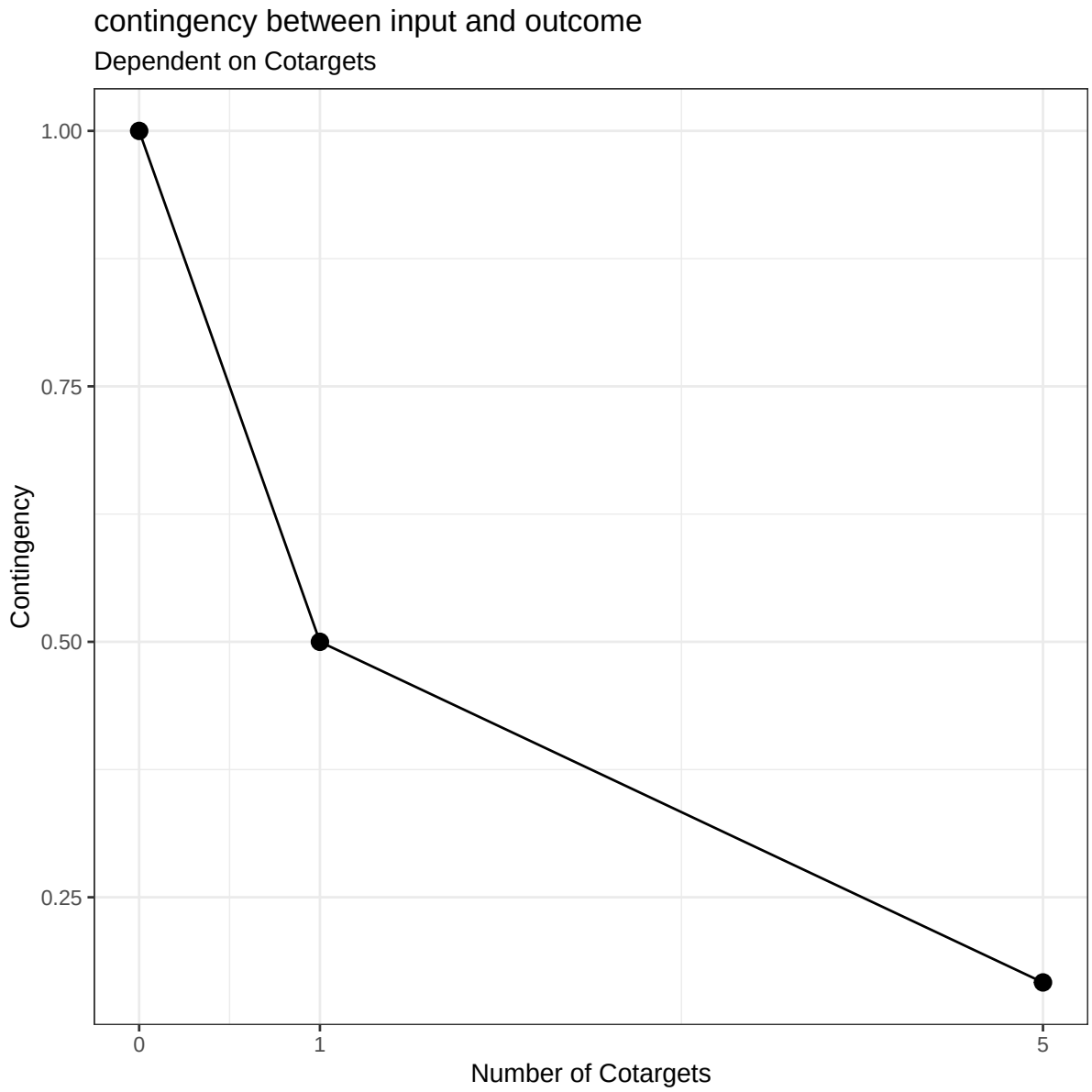


Figure 5

Plot for the transformation t1

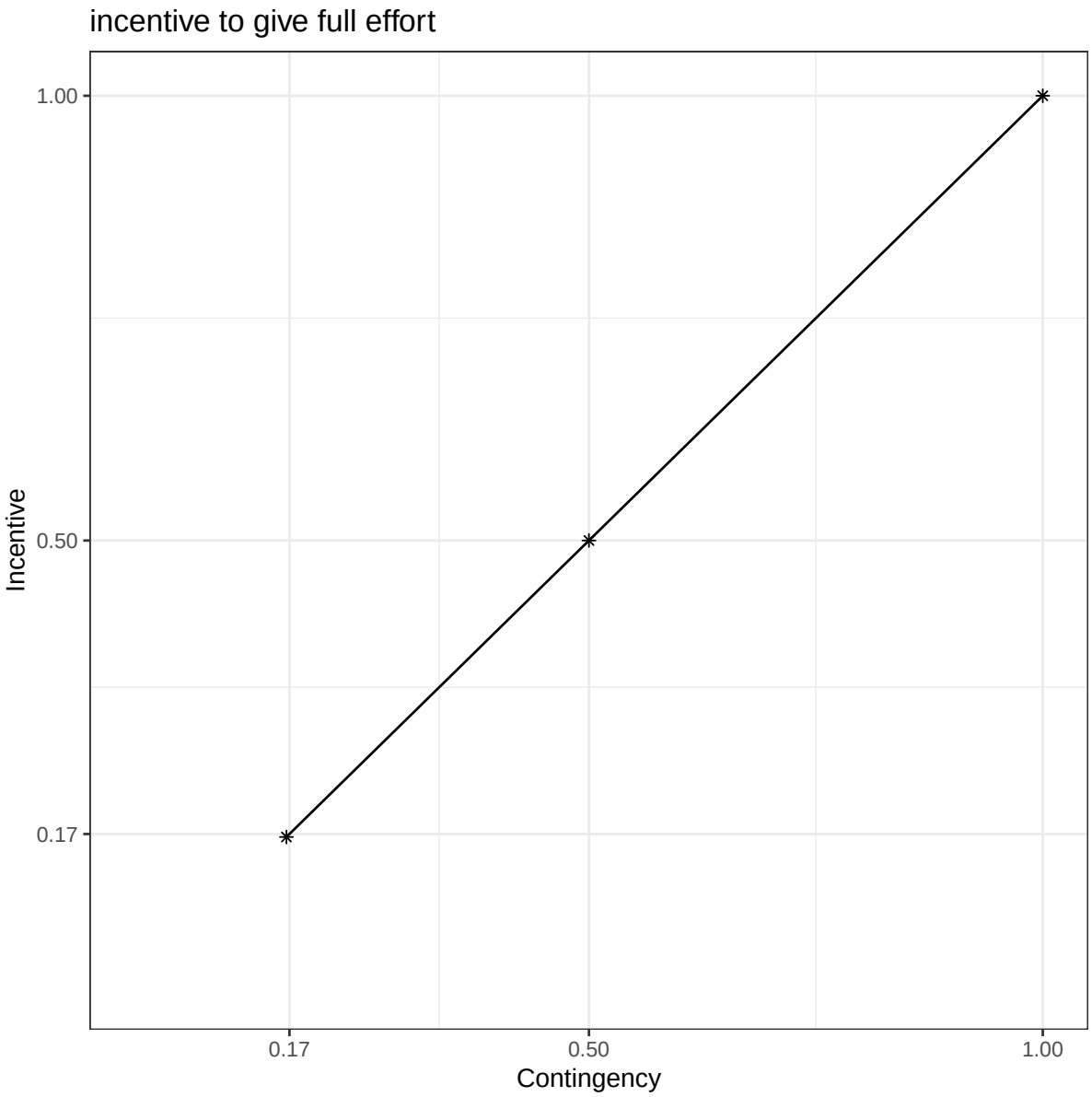


Figure 6

Plot for the relationship p3

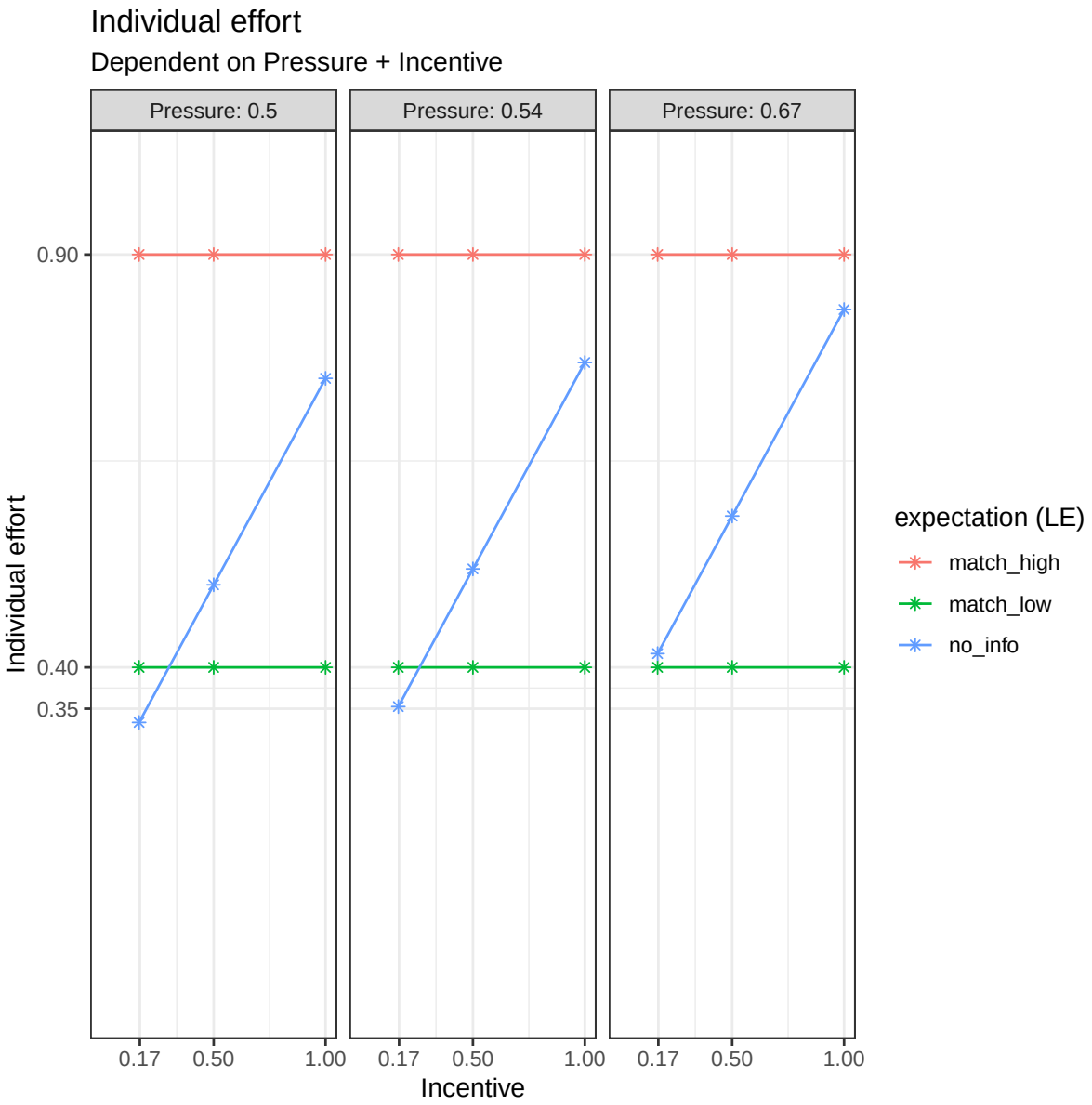


Figure 7

Plot for the relationship t2

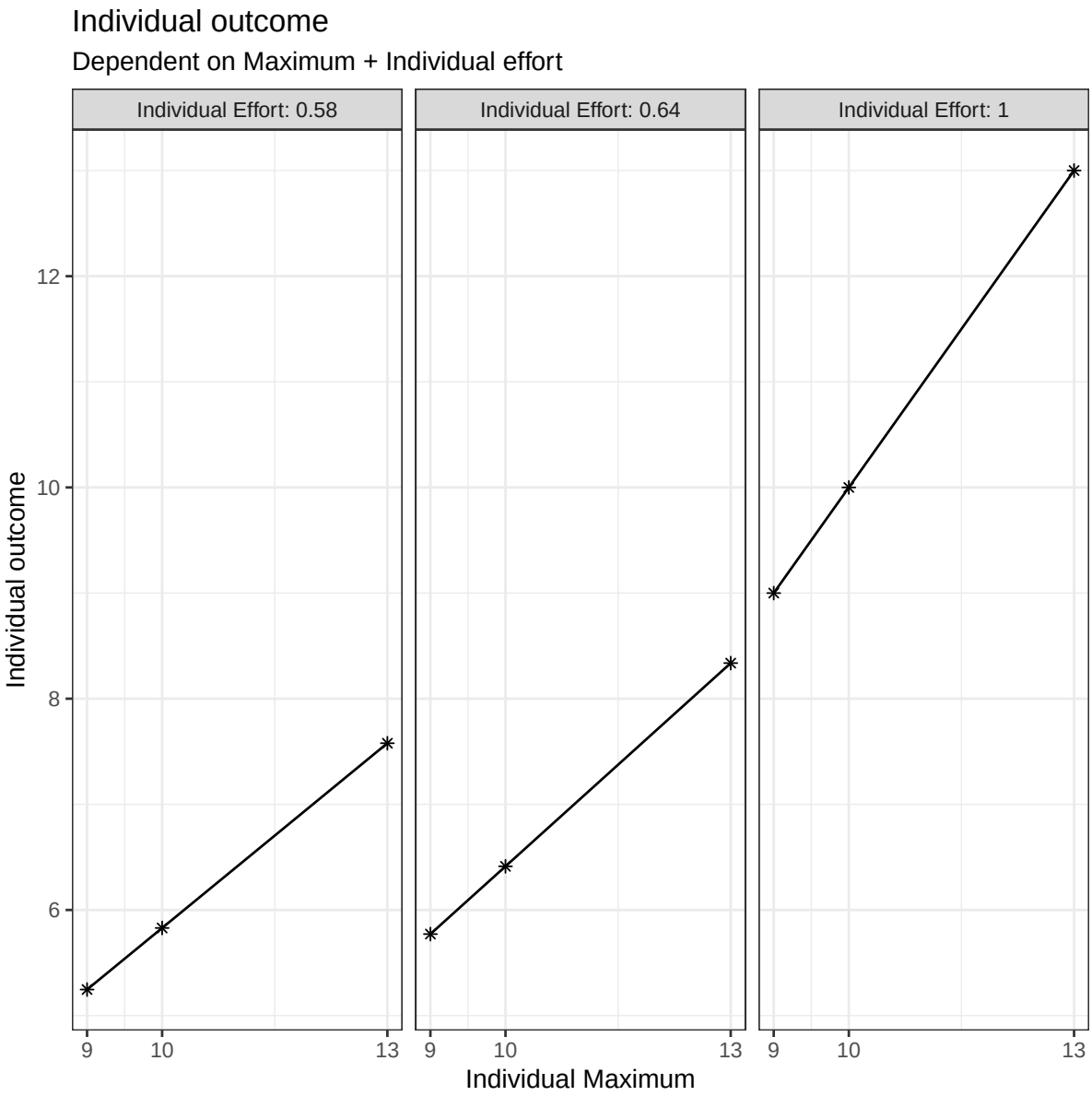


Figure 8

Boxplot for the replication of Latané

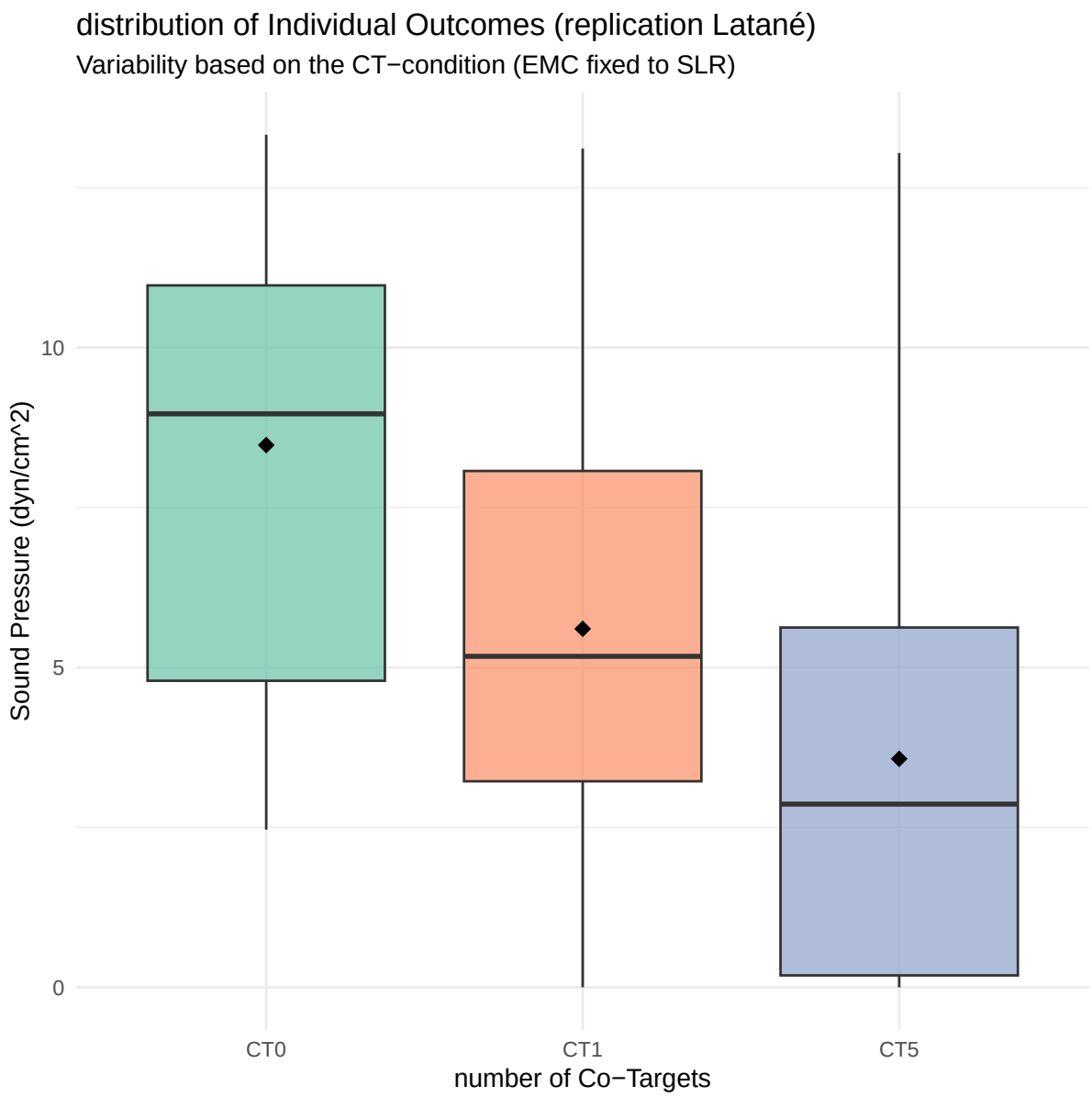


Figure 9

Boxplot for the replication of Jackson & Harkins

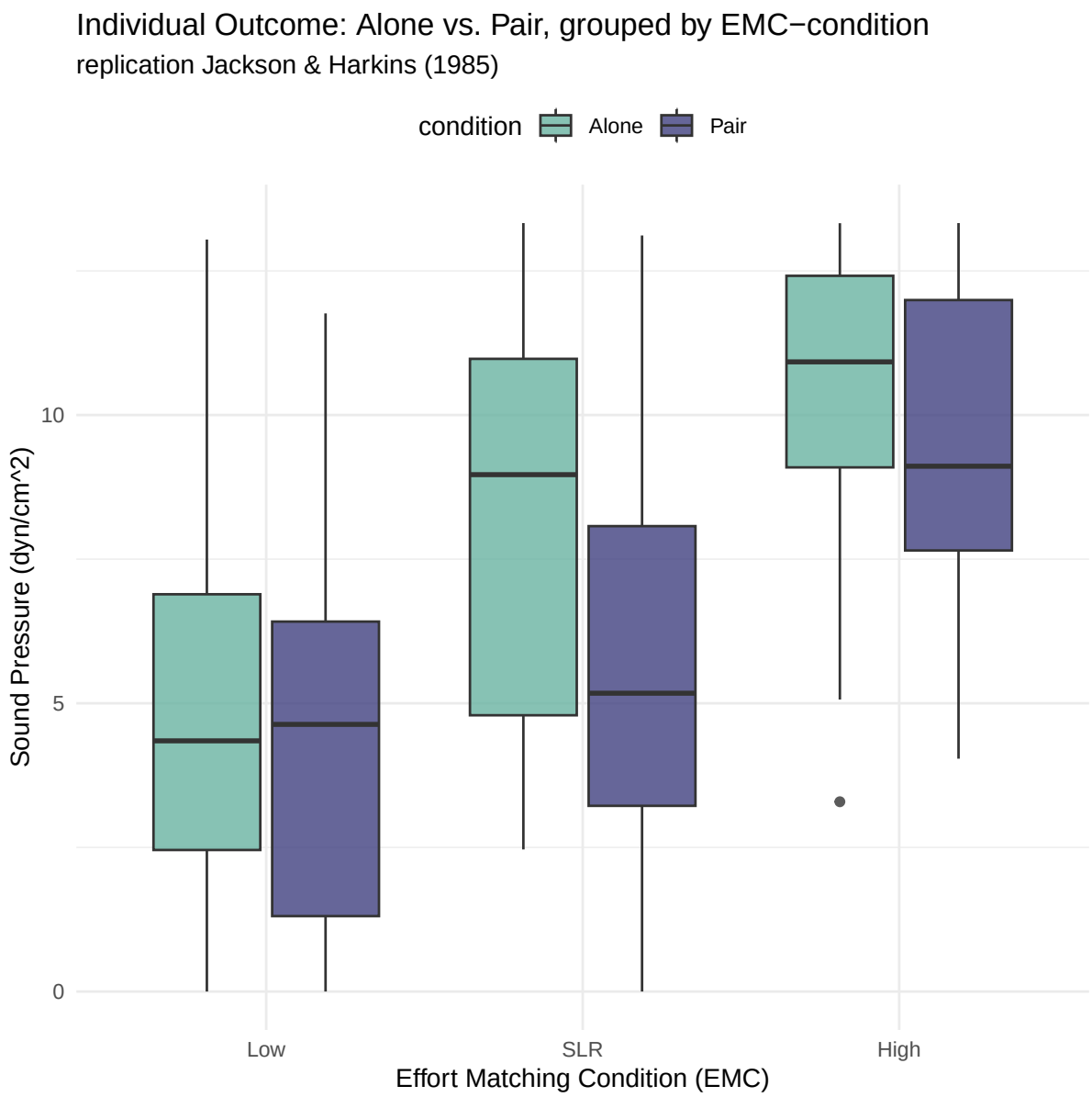


Figure 10

Final plot of the simulation

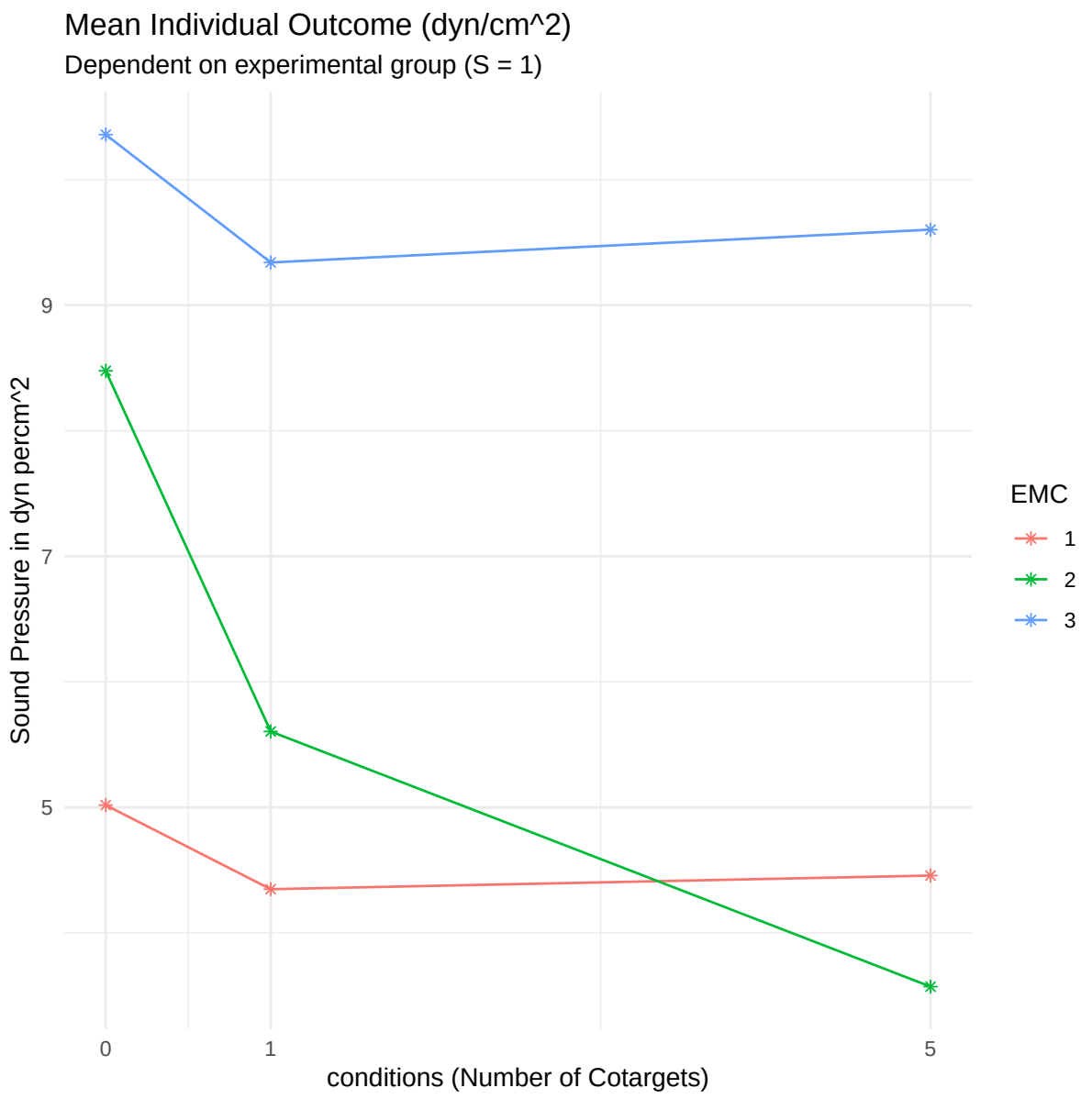
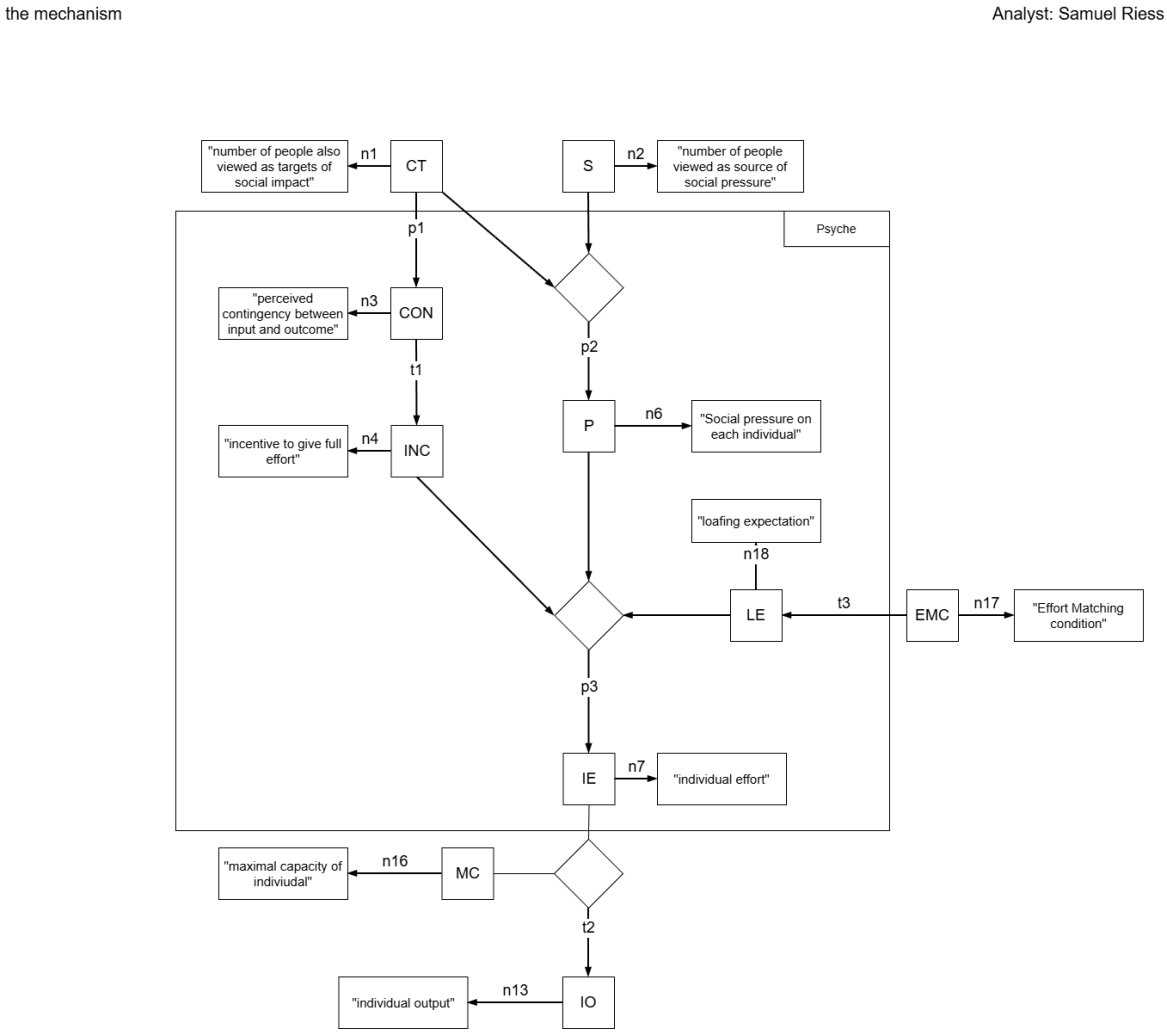


Figure 11

Structure of the loafing mechanism without Fallacies